Higher National Diploma in Engineering (Civil)

3rd Year, Second Semester Examination – 2021

CE3120 - Materials Engineering

Instructions for Candidates:
Answer for **four** questions only

No. of questions : 06
No. of pages : 03
Time : 02 hours

01.

(i) Draw a neat sketch of stress vs strain curves of ductile and brittle material. Explain the difference between ductility and brittleness. (05 marks)

(ii) Explain elastic and plastic deformation on brittle and ductile materials with respect to above stress-strain graph and explain it with real world examples. (06 marks)

(iii) Define the terms "Fatigue" and "Creep". (04 marks)

(iv) Sketch and label a typical creep curve (Stain vs time) for a hypothetical material and discuss the behavior of a material at high temperatures using the graph. (05 marks)

(v) Explain in brief the fatigue crack growth mechanism. (05 marks)

02.

(i) What is an alloy? State the difference between ferrous and non-ferrous alloys. (05 marks)

(ii) Draw a neat sketch of a phase diagram of binary isomorphus system. Explain how you would obtain the compositions of each component in solid (α) and liquid phases. (07 marks)

(iii) Sketch Fe-C diagram and mark peretectic, eutectic and eutectoid points and write the reaction happened on each point. (08 marks)

(iv) State five methods used for prevention of corrosion in metals. (05 marks)

Damaging the polymer due to its service good or bad.
- Ozone
- Heating
- Chemical reaction
- Radiont.

2.66 x 10^3

03.

(i) Define the following terms

(a) Crystal lattice

(b) Unit cell

(c) Atomic packing factor (06 marks)

(ii) Calculate the atomic packing factor of following unit cells, assuming the atoms to be hard spheres.

(a) Face Centered Cubic (FCC)

(b) Body Centered Cubic (BCC) (08 marks)

(iii) Explain the relationship between atomic packing factor and density. (06 marks)

(iv) The Vanadium has a BCC crystal structure. A density of 5.96 g/cm^3, and an atomic weight of 50.9 g/mol. Compute the radius of Vanadium atom. (05 marks)

Domestic goods.
Thermal insulator
heat insulator
Decorative Aurnoments.
Cutters.

04.

(i) Define Hardness of water. (04 Marks)

(ii) Differentiate Temporary hardness and Permanent hardness of water. (06 Marks)

(iii) Explain disadvantages of water hardness in civil engineering context. (05 Marks)

(iv) A sample of water sample contains,

$Mg(HCO_3)_2$ = 73mg/L, $Ca(HCO_3)_2$ = 162mg/L, $MgCl_2$ = 95mg/L, $CaSO_4$ =136mg/L

Calculate,

(a) Temporary hardness

(b) Total hardness (07 Marks)

(v) Explain what do you understand by Boiler Blow down. (03 marks)

05.

(i) State four properties of ceramics. (04 marks)

(ii) State five applications of ceramics. (05 marks)

(iii) What do you understand by degradation of polymers? State four factors of polymer degradation. (05 marks)

(iv) Classify the composite materials. metal + nonmetal. (05 marks)

(v) Define the following terms, (05 marks)

(a) Thermal conductivity

(b) Intrinsic semiconductors

(c) Glass transition temperature (06 marks)

06.

(i) Define the terms "Admixtures" and "Plasticizers" (06 marks)

(ii) Describe five selection criteria for admixture (05 marks)

(iii) Name five primary types of Portland cement? Briefly discusses the function of each type. (07 marks)

(iv) Describe the hydration process of ordinary Portland cement mentioning the chemical reaction. (07 marks)